Code No: 136FQ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February/March - 2022
NON-CONVENTIONAL ENERGY SOURCES
(Common to CE, EEE, ME, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
- - -

1.a) Differentiate between renewable and non renewable energy sources.
b) Explain the different types of renewable energy systems. [8+7]

2.a) Write a short notes on concept of hybrid systems.
b) Describe about Global Climate change. [8+7]

3.a) Define solar constant. What are the reasons for variation in solar radiation reaching the earth and that received outside the earth atmosphere?
b) Differentiate between Beam and diffuse radiation. [9+6]

4.a) Explain the working principle of PV System with neat diagram.
b) Write note on suns declination and Hour angle. [9+6]

5. Explain the working of horizontal axis turbines with a neat sketch and mention their advantages and disadvantages. [15]

6. Write short notes on:
a) Wind energy
b) Darrius roton. [7+8]

7. Explain the process of production of biogas from biomass. What are the advantages of anaerobic digestion of biomass? [15]

8.a) What are the advantages and disadvantages of geothermal energy conversion?
b) Describe vapour dominated hydrothermal (convective) power plant with neat sketch. [8+7]

---ooOoo---

used paper for/mar-2022 jntuh